

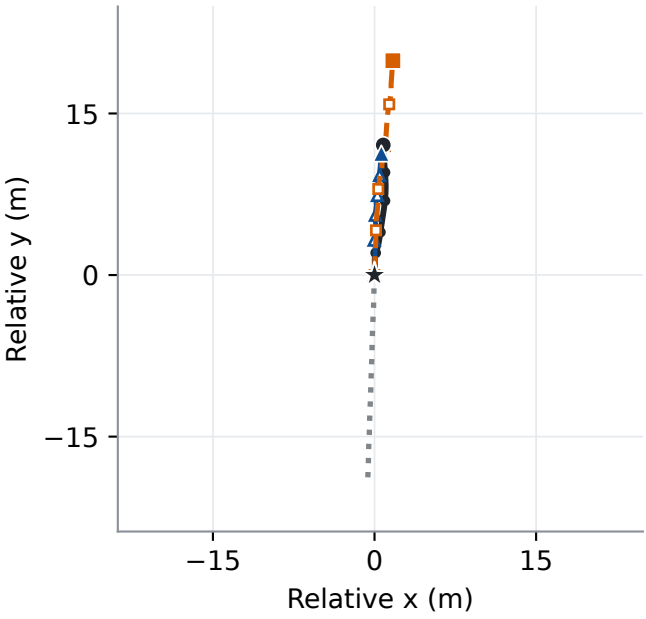
Where inherited QGNN outperforms the strong GNN

Selected cases | V_select, SNR 20 dB | QGNN checkpoint 75; GNN checkpoint 97

..... Past estimate —●— Future truth —▲— QGNN —■— GNN ★ Present target

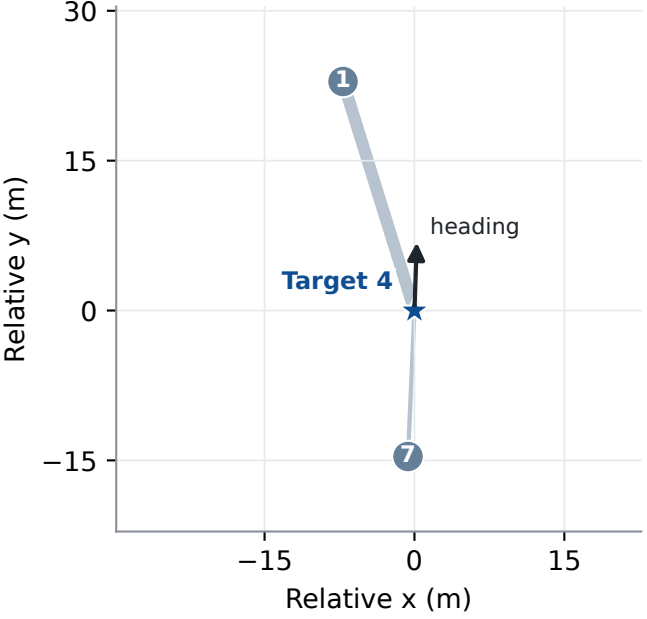
TARGET TRAJECTORY

Case 131 | target slot 4
ADE Q 1.01 / G 4.47 m; FDE Q 0.81 / G 7.88 m



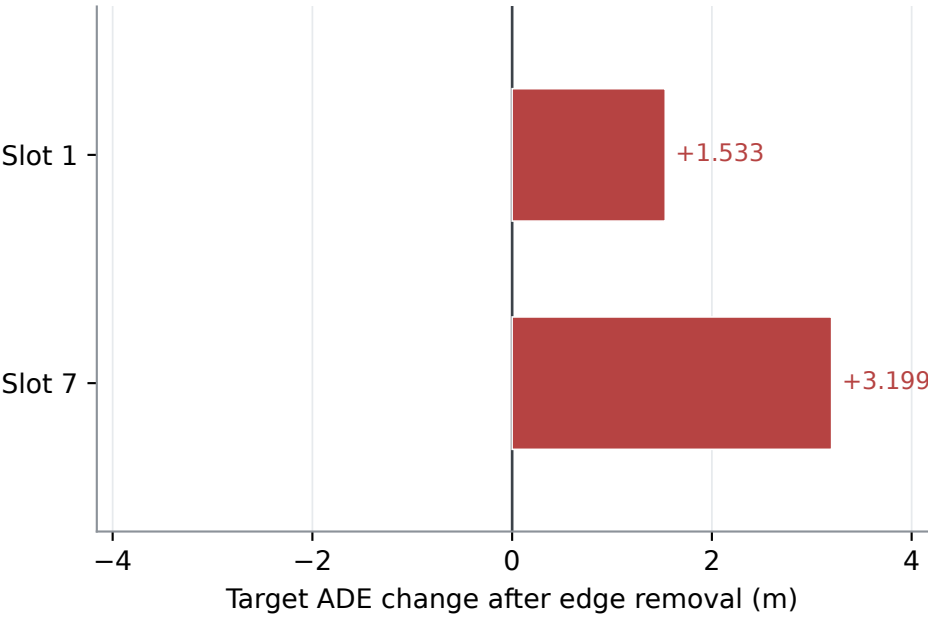
NEIGHBOR CONTEXT

Direct neighbors seen in history
Positions now; width = layer-2 contribution proxy

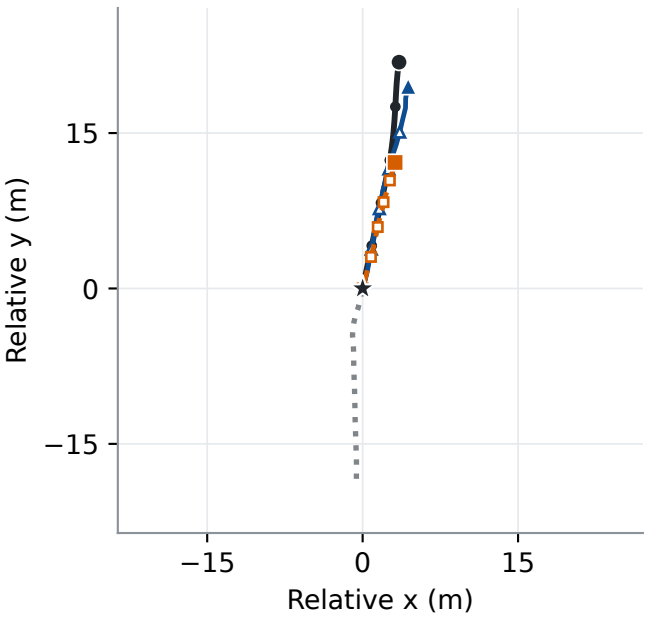


FIXED-MODEL INTERVENTION

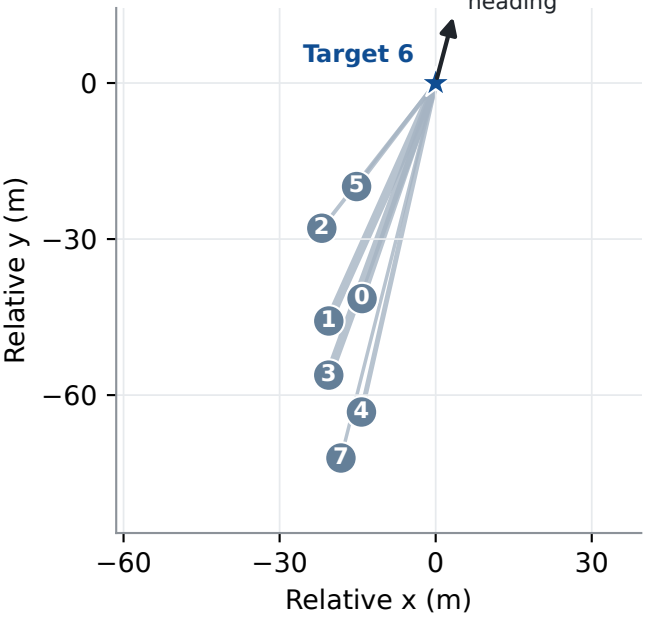
Remove one neighbor-to-target edge
Fixed QGNN; no retraining



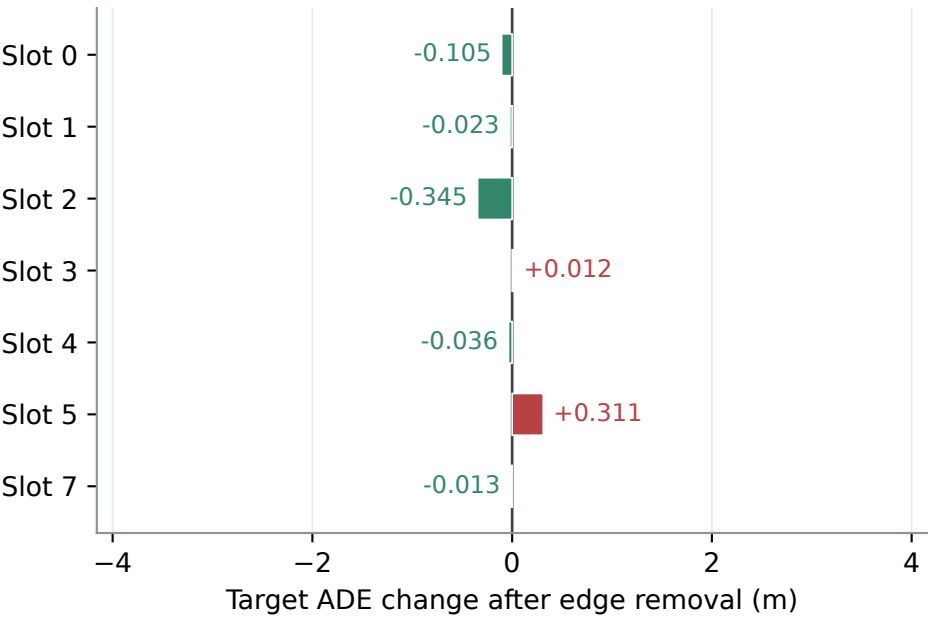
Case 32 | target slot 6
ADE Q 1.22 / G 4.14 m; FDE Q 2.50 / G 9.69 m



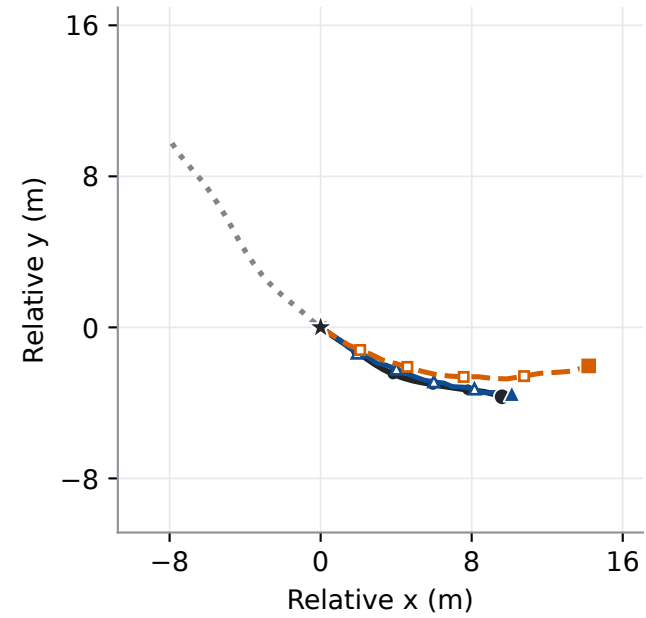
Direct neighbors seen in history
Positions now; width = layer-2 contribution proxy



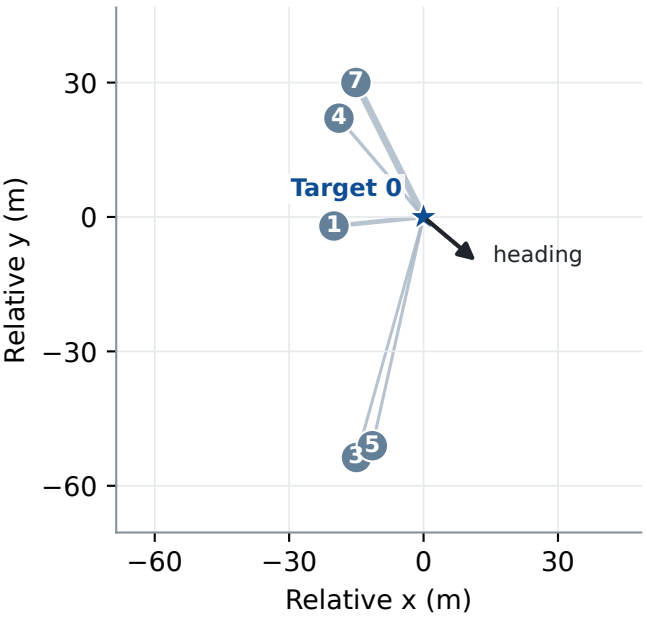
Remove one neighbor-to-target edge
Fixed QGNN; no retraining



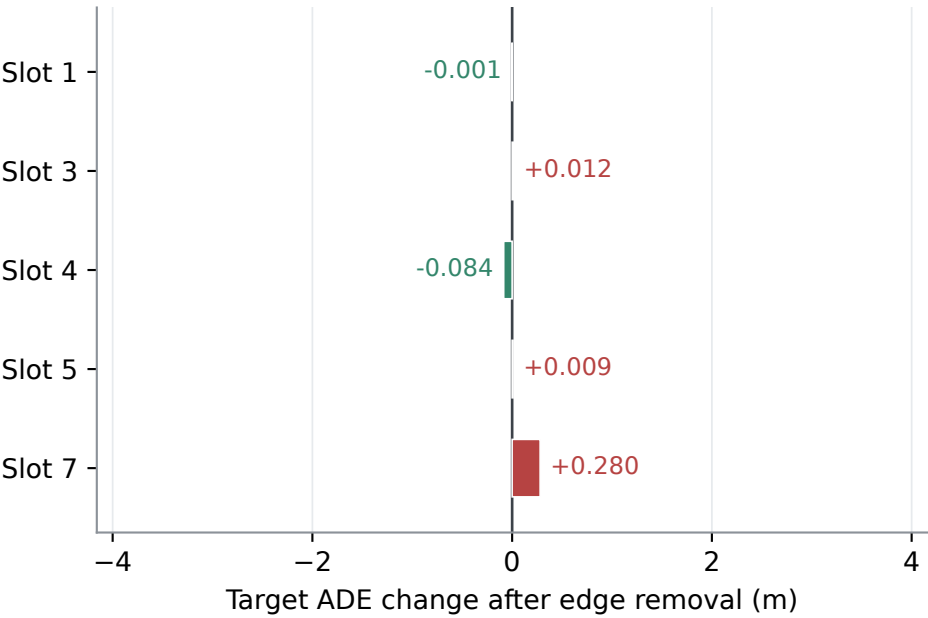
Case 262 | target slot 0
ADE Q 0.23 / G 1.73 m; FDE Q 0.54 / G 4.87 m



Direct neighbors seen in history
Positions now; width = layer-2 contribution proxy



Remove one neighbor-to-target edge
Fixed QGNN; no retraining



Coordinates are translated to the present target; spatial axes have equal meter scales. Neighbor line widths encode history-mean layer-2 contribution norms (proxy only).

Green / negative: removal improves target ADE. Red / positive: removal worsens it. Fixed-model edge removal is a sensitivity diagnostic, not evidence of real-world causal importance.